Journal of VLSI Design Tools and Technology
Volume 3, Issue 1, ISSN: 2249- 474X

A New High Speed Low Power 1 Bit Full Adder

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Abstract

This paper proposes a low power, low area and high speed 1-bit full adder compared to earlier 1bit full adder designs. The adder comprises of eight transistors with some modifications over the other existing design, so that it can invariably be used in the device with least nano device dimensions. The proposed design exhibits lower threshold loss and works faithfully in sub-threshold regime with appreciable voltage swing at lower power supply of 0.5 V. The simulation has been carried out using Cadence Virtuoso Spectre at 45 nm technology.

Keywords: *design and simulation, full adder, low power, nanoscale, sub-threshold*

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INTRODUCTION

The urge for reduction of device dimension, so as to compute more function per unit area has invited alarming elevation in power consumption due to uncontrolled leakage current in nano scaled devices. This in turn deteriorates the stand-by operating time of small hand-held battery operated devices. High power consuming designs increases the need for cooling arrangement and decreased reliability for high density systems. In this nano meter regime, compact design of adder circuit is a challenging task. Several issues need to be considered in designing adder circuits. To name some of the most crucial aspects of full adders are: power consumption, performance, area, noise immunity, regularity and good driving capability. Several ideas have been proposed in order to reduce total transistor count, power and area. Some of them suffer from serious problem of threshold loss that causes non-full-swing of the out-put. To minimize some of the hindrances mentioned above a new scheme in which all the transistors are forced to operate in sub-threshold regime is adopted. Sub-threshold current of MOS comes into picture when $V_{GS} < V_{TH}$. This current flows due to minority carriers from drain to source in weak inversion mode. In standard MOS affair, this current is waste/ parasitic. But if the supply voltage (V_{DD}) is lowered below V_{TH}, the circuit can be operated using the sub-threshold current with ultra-low power consumption [1–3]. Moreover, as the circuit is used in sub-threshold regime, there is no need for additional V_{DD} supply in the design. This simplifies the design further. Our proposed design exhibits larger delay compared to simple 8T based 1 bit full adder, but lesser delay to that of 9T 1 bit full adder. The proposed design saves more power compared to both of the designs mentioned above. Thus, it can be proven more efficient out of the three designs.

PRIOR WORK

1 Bit Full Adder with Eight Transistors

A 1 bit full adder constructed of eight transistors is shown in Figure1. The sum output is obtained by cascading exclusive OR-ing of the three inputs [4, 5]. Alternately, C_{OUT} is implemented using 2T multiplexer. From Figure1, it is clearly evident that at least two stage delay is required to obtain both Sum and Carry output. To achieve a full swing at the output which is highly influenced by threshold loss, aspect ratio of the transistors has to be chosen wisely. However, the circuit suffers from a serious problem of threshold loss whenever input C=’0’. Due to this, there is a major degradation at the output for some of the input vector combination. The magnitude of the output may even be lowered to a value where output may become equal to V_{DD}/ 2. The out-put waveform is shown in Figure 2.

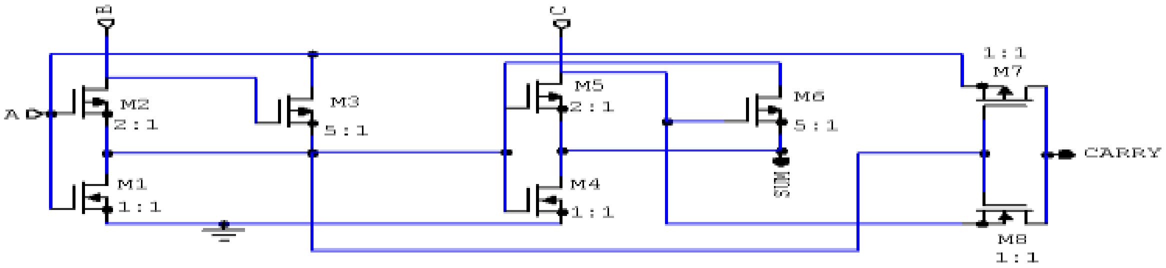

Fig.1: *1 Bit Full Adder with Eight Transistors.*

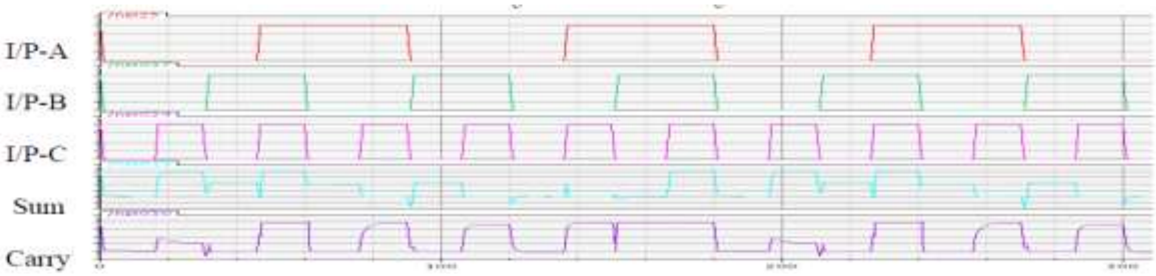

Fig. 2: *Input-Output Waveforms of 1 Bit Full Adder with Eight Transistors.*

1 bit Full Adder with Nine Transistors

To overcome the above mentioned problem, a new idea has been proposed in [8]. In this adder (Figure 3), an additional NMOS has been added to overcome threshold loss problem. This is because during the input combinations of ABC = 000, 010 and 110, two transistors turns on simultaneously. As a result of which a combined parallel effect of reduced resistance takes place which in turn deteriorates the out-put. The addition of an extra transistor eliminates this problem. Now with in-puts 000, 010 and 110, the XOR of first stage gives a complete logic 1 which enables the extra added transistor, giving a complete 0 for C_{IN} = 0. Thus, the problem for full swing at the out-put gets eliminated. The out-put waveform is shown in Figure 4

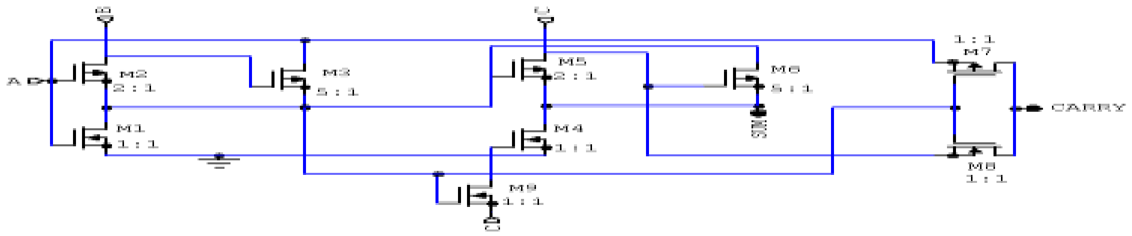

Fig. 3: *1 Bit Full Adder with Nine Transistors*.

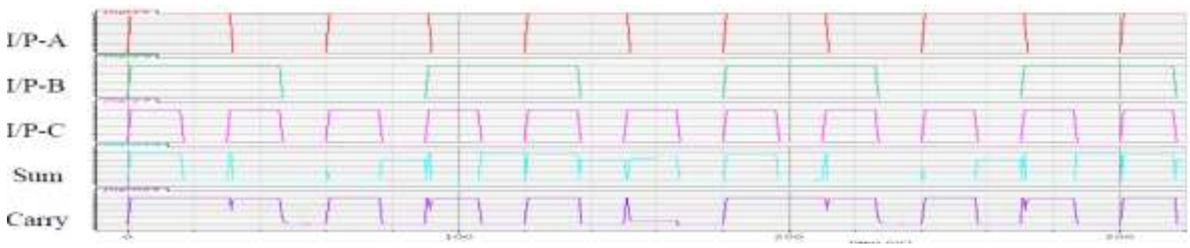

Fig. 4: *Input-Output Waveforms of 1 Bit Full Adder with Nine Transistors.*

PROPOSED DESIGN OF 1 BIT FULL ADDER AND ITS WORKING

Circuit Description

The design proposed in [8] works satisfactorily. But some small refinements can be done in the existing design to achieve better power and delay and area response. In our design, we have replaced the additional transistor (M9) by a capacitor of small value of 75 aF to drive transistor M4. This has eliminated the need for fabrication of an additional transistor M9. The capacitor is quite simple to fabricate using metal-insulator-metal structure compared to transistor. The aspect ratio of all the transistors has to be chosen wisely in order to have a full swing out-put. Design and simulation results of the proposed adder are shown in Figures 5 and 6, respectively.

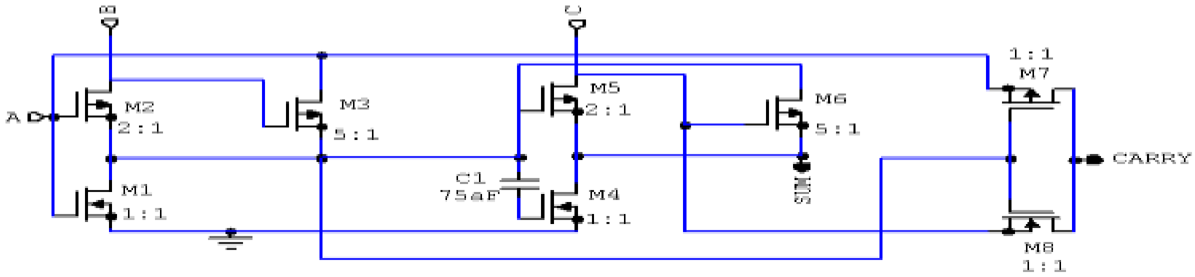

Fig. 5: *Proposed 1-Bit Full Adder.*

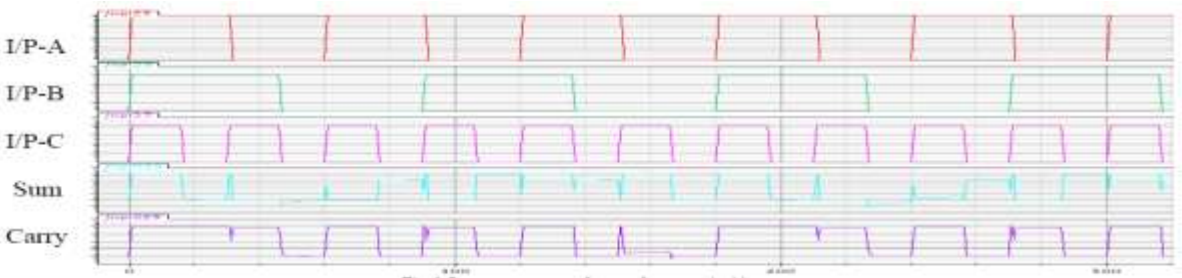

Fig. 6: *Input-Output Waveforms of Proposed Adder.*

Working of Circuit

Whenever a voltage is applied to a capacitor, a differential voltage arises across the capacitor. The delay and voltage differential produced across the capacitor keeps the NMOS transistor at negatively biased gate. This eliminates the problem of reduced parallel resistance due to which two of the transistors used to turn on at the same time for the combinations of 000, 010 and 110, which in turn helps in reducing the threshold loss resulting in full swing of the output. The delay offered by the additional transistor in existing adder design is 89.261 pico sec to drive M4 transistor is more in comparison to the capacitor delay of 88.195 pico sec in the existing design. The excess delay of 1.066 pico sec in the existing design, in turn retards the Sum output resulting in a slow response device. Furthermore, the power consumption of the existing adder is more than that of our proposed design. This is because, the source terminal of the additional transistor in existing adder is tied directly to C input. Thus, whenever input A and C are opposite in logic level to each other, there exist a leakage path for the input signals to dissipate into each other. This in turn elevates the power consumption level of the logic circuit. The comparison of power and delay of the proposed design with existing two designs has been tabulated in Table 1 and 2.

SIMULATION AND RESULT

The design has been simulated using Cadence Virtuoso Spectra at 45 nm TSMC technology. The circuit works quite satisfactorily even at 0.2 volt range. A parametric simulation of power and delay for input voltage ranging from 0.1 to 0.5 V has been carried out and results have been tabulated in Table 1 and Table 2. To establish an impartial environment, all the designs have been tested under same input pattern vector with exactly the same time period. All the three designs have been compared in terms of threshold loss. The values have been tabulated in Table 3. Performance of 8T, 9T and proposed full adder in terms of Threshold loss is given in Table 4. All the data comparison tabulated above has been plotted in graph form and shown from Figures 7–13.

Table 1: *(Power Consumption Comparison for 8T, 9T and Proposed Full Adder).*

A	B	C	8T 1bit Full Adder												Time Period us
			Power						Total Average Power						
			.1V	.2V	.3V	.4V	.5V	1V	.1V	.2V	.3V	.4V	.5V	1V	
0	0	0	75f	305f	500f	1.8p	7.49p	1p							0 – 15
0	0	1	117f	423f	618f	5.1p	91.7p	14p							16–30
0	1	0	202f	10.6p	302p	5.4n	71.1n	56p	1.22p	69.43p	2.47n	43.27n	343.8n	419n	30-45
0	1	1	267f	15.5p	580p	9.6n	51.9n	66p							105-120
1	0	0	197f	6.78p	275p	3.3n	55.6n	74p							60-75
1	0	1	234f	7.67p	423p	7.2n	60.9n	79p							75-90
1	1	0	180f	2.2p	212p	2.1n	50.7n	60p							150-165
1	1	1	209f	3.9p	425p	6.2n	41.1n	226n							45-60
A	B	C	9T 1bit Full Adder												Time Period us
			.1V	.2V	.3V	.4V	.5V	1V	.1V	.2V	.3V	.4V	.5V	1V	
0	0	0	79f	293f	370f	1.5p	6.50p	.9p							0 – 15
0	0	1	119f	417f	600f	4.8p	80.7p	13.2p							16–30
0	1	0	212f	9.3p	292p	4.3n	44.1n	53p	1.3p	55.18p	1.91n	30.07n	223.4n	357n	30-45
0	1	1	273f	12.5p	512p	7.4n	41.9n	62p							105-120
1	0	0	200f	5.38p	261p	3.1n	30.6n	71p							60-75
1	0	1	240f	5.23p	403p	6.9n	41.9n	77p							75-90
1	1	0	190f	2.1p	198p	2.1n	39.7n	56p							150-165
1	1	1	213f	3.2p	397p	5.1n	30.1n	221n							45-60
A	B	C	Proposed 8T 1 bit full adder												Time Period us
			.1V	.2V	.3V	.4V	.5V	1V	.1V	.2V	.3V	.4V	.5V	1V	
0	0	0	62f	290f	364f	1.5p	5.90p	.63p							0 – 15
0	0	1	100f	409f	589f	4.8p	78.8p	12.1p							16–30
0	1	0	168f	8.35p	280p	4.3n	42.6n	51.1p	1.12p	52.68p	1.87n	29.60n	221n	350n	30-45
0	1	1	227f	11.9p	501p	7.4n	36.1n	60p							105-120
1	0	0	155f	5.00p	225p	3.1n	29.3n	69.1p							60-75
1	0	1	201f	5.12p	332p	6.9n	39.9n	75.6p							75-90
1	1	0	127f	1.91p	160p	2.1n	37.6n	55.1p							150-165
1	1	1	175f	2.74p	319p	5.1n	28.1n	215n							45-60

A = Period of 90 uS with 50% Duty cycle.B = Period of 60 uS with 50% Duty cycle.C = Period of 30 uS with 50% Duty cycle.

Table 2: *(Delay Comparison Results for 8T, 9T And Proposed Full Adder).*

IP Voltage		Sum Delay			Carry Delay		
		8T Adder	9T Adder	Proposed Adder	8T Adder	9T Adder	Proposed Adder
0.1v	Rise Time	8.28u	23u	15u	5.3u	5.7u	5.6u
	Fall Time	911n	925n	915n	850n	855n	851n
0.2v	Rise Time	1.75u	1.7u	1.5u	1u	1.3u	1.06u
	Fall Time	858n	869n	853n	628n	833n	823n
0.3v	Rise Time	460n	835n	827n	166n	825n	810n
	Fall Time	559n	850n	843n	544n	823n	818n
0.4v	Rise Time	400n	766n	758n	73n	808n	800n
	Fall Time	400n	839n	833n	408n	820n	814n
0.5v	Rise Time	517n	755n	748n	500n	788n	789n
	Fall Time	656n	832n	827n	600n	812n	800n
1.0v	Rise Time	310n	400n	388n	236n	413n	403n
	Fall Time	400n	492n	480n	325n	593n	589n

Table 3: *(Average Delay Comparison for Sum And Carry Output with 8T, 9T And Proposed Full Adder).*

Designs	Sum	Carry
9T Adder	**27.456 uS (RT)** **4.807 uS (FT)**	**9.834 uS (RT)** **4.736 uS (FT)**
Total Delay	**32.263 uS**	**14.57 uS**
Proposed Adder	**19.221 uS (RT)** **4.451 uS (FT)**	**9.462 uS (RT)** **4.695 uS (FT)**
Total Delay	**23.672 uS**	**14.157 uS**
% Saving	***26.62***	***2.83***

Table 4: *Performance of 8T, 9T and Proposed Full Adder in Terms of Threshold Loss.*

A	B	C	Sum 8T [7]	Sum 9T [8]	Proposed Adder
0	0	0	<< Vtp	<< Vtp	<< Vtp
0	0	1	1	1	1
0	1	0	<30% of logic 0	1	1
0	1	1	0	0	0
1	0	0	<30% of logic 1	1	1
1	0	1	0	0	0
1	1	0	<< Vtp	<< Vtp	<< Vtp
1	1	1	1	1	1

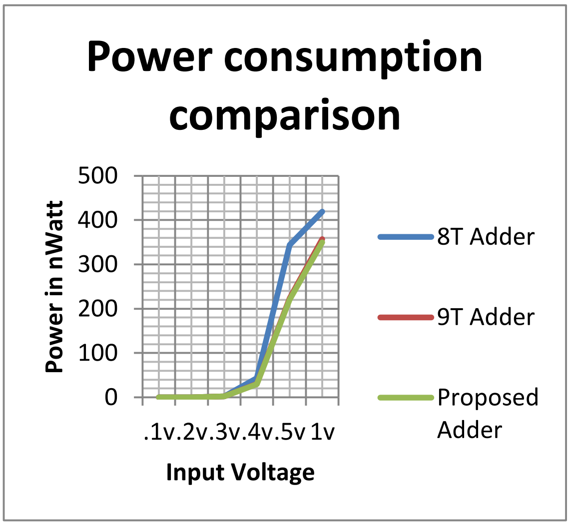

Fig. 7: *(Power Consumption Comparison).*

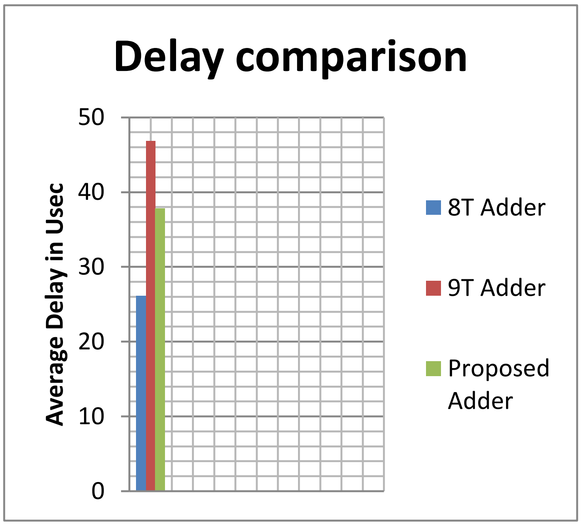

Fig. 8: *(Delay Comparison).*

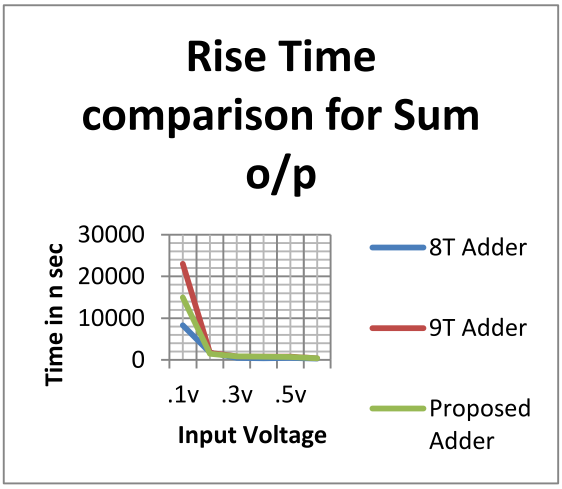

Fig. 9: *(Rise Time Comparison of Sum o/p).*

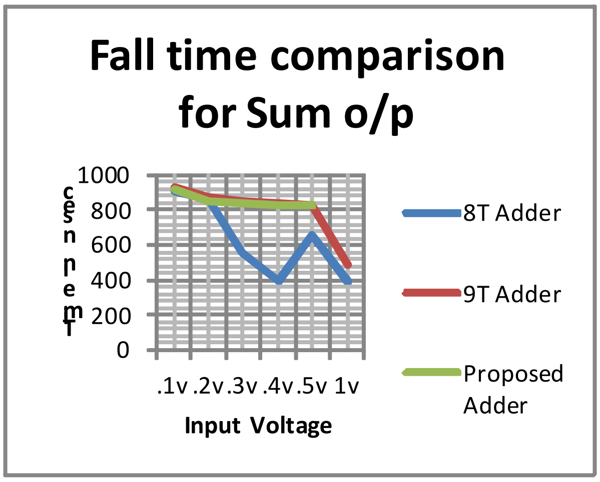

Fig. 10: *(Fall Time Comparison of Sum o/p).*

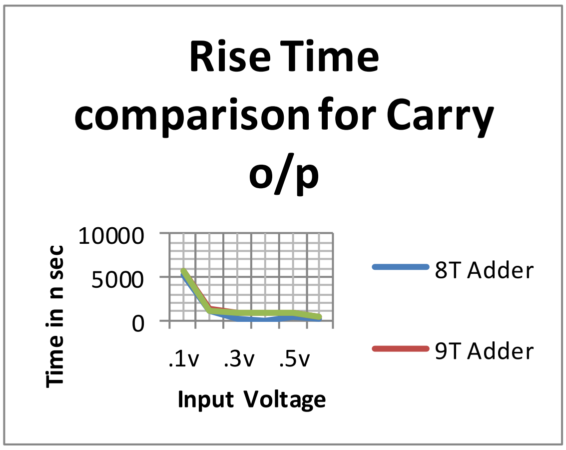

Fig. 11:*(Rise Time Comparison of Carry o/p).*

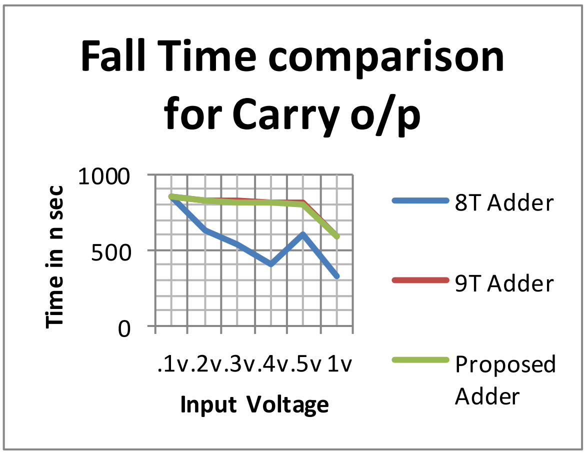

Fig. 12:*(Fall Time Comparison of Carry o/p).*

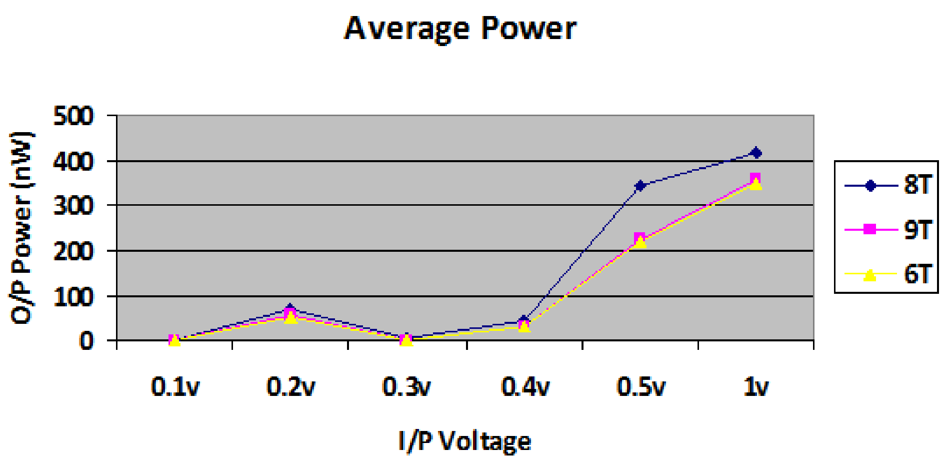

Fig. 13: *(Average Power Consumption Plot).*

CONCLUSION

Authors proposed design of full adder circuit performs efficiently in sub-threshold regime. There is significant savings in delay and power of the proposed design compared to earlier design. Design and simulation have been performed in submicron level. The efficiency achieved for the factors noted above makes the design a viable option to be used in ultra-low power application in submicron level.

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